Data Analytics Internship Portfolio

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This document serves as the comprehensive, consolidated portfolio detailing the technical workflows, methodologies, and business insights extracted over the first two weeks of the DecodeLabs Data Analytics Internship. It showcases the transition from structural data sanitization to programmatic exploratory data analysis.

Week 1

Project 1: Data Cleaning & Preprocessing

Microsoft Excel | Data Sanitization | Structural Formatting

The foundational phase of the internship focused on transforming raw, unstructured enterprise data into a reliable, analytical-ready dataset. The integrity of any predictive model or dashboard relies fundamentally on the hygiene of the input data.

1. Objective & Scope

To audit, clean, and standardize a messy sales dataset, removing inconsistencies, correcting data formats, and establishing a robust "single source of truth" for downstream analytical processes.

2. Execution Methodology

- **Deduplication:** Executed a comprehensive scan to identify and remove redundant row entries, ensuring transaction volumes and revenue metrics remain uninflated.
- **Missing Value Imputation:** Addressed null values logically. Missing nominal data was categorized appropriately, and numerical gaps were treated without skewing the variance.
- **Type Casting & Formatting:** Standardized temporal fields (Dates) and financial fields (Currency) into consistent formats, rectifying alignment and trailing/leading space issues.
- **Integrity Validation:** Conducted a final cross-reference check to ensure no anomalies persisted across the `UnitPrice`, `Quantity`, and `TotalPrice` columns.

Week 2

Project 2: Exploratory Data Analysis (EDA) & Programmatic Profiling

Python Pandas Matplotlib Seaborn

Building upon the sanitized dataset from Week 1, the objective shifted toward algorithmic data discovery. This phase utilized Python's analytical libraries to programmatically uncover underlying patterns, hidden correlations, and significant outliers within the enterprise data.

1. Statistical Summary & Central Tendencies

Utilizing Pandas' `describe()` framework, numerical matrices were profiled to calculate foundational metrics (Mean, Median, Standard Deviation, and Quartiles). This provided an immediate mathematical baseline of the dataset's geometry and distribution.

2. Unmasking Outliers (Anomaly Detection)

A statistical Boxplot visualization was generated using Seaborn to map the variability of the `TotalPrice` feature.

> **Finding:** The plot successfully isolated significant data points beyond the upper whisker. In a business context, these are not "noise" or entry errors, but rather critical "signals" representing high-value VIP customer purchases or bulk B2B transactions that require dedicated retention strategies.

3. Correlation vs. Causation Analysis

A Pearson Correlation Coefficient was calculated programmatically to investigate the relationship between product cost and consumer purchasing volume.

- **Coefficient Result:** $r \approx 0.028$
- **Analytical Insight:** The result indicates near-zero linear correlation. Customers purchase the quantities they require irrespective of the unit price variance. This lack of dependency was further visually validated via a generated Scatter Plot.

4. Categorical Performance & Business Recommendations

Through GroupBy aggregation on the `Product` axis, a Bar Chart was constructed to rank inventory demand. The visual evidence clearly identified the top-performing categories (e.g., Printers and Chairs), serving as a direct compass for inventory procurement forecasting and targeted marketing allocations.